

Assignment 3

Problem Statement: Fine-tune GPT or GPT-2 for creative story generation.

Introduction

Fine-tuning is a technique used in Natural Language Processing (NLP) to adapt a pre-trained language model, such as GPT or GPT-2, to a specific task or domain. While these models are trained on large-scale general datasets, fine-tuning allows them to specialize in generating content tailored to a particular style, theme, or application.

In this assignment, a GPT/GPT-2 model is fine-tuned for **creative story generation**, enabling it to produce engaging, coherent, and context-aware narratives. This process enhances the model's ability to generate imaginative stories with improved structure, tone, and creativity compared to generic outputs.

Objective

- To understand the concept of fine-tuning in deep learning and NLP.
- To fine-tune a pre-trained GPT/GPT-2 model for story generation.
- To train the model on a dataset of creative stories.
- To evaluate the performance of the fine-tuned model in generating coherent and creative narratives.
- To compare the outputs of the base model and fine-tuned model.

Theory

1. Pre-trained Language Models

Models like GPT and GPT-2 are based on the **Transformer architecture**, which uses self-attention mechanisms to understand context in text. These models are trained on massive datasets and can generate human-like text.

However, their outputs are often generic unless adapted to a specific domain using fine-tuning.

2. Fine-Tuning

Fine-tuning involves taking a pre-trained model and training it further on a smaller, domain-specific dataset.

Steps in Fine-Tuning:

1. Select a pre-trained model (e.g., GPT-2)
2. Prepare a dataset (collection of stories)
3. Tokenize the dataset
4. Train the model on the dataset
5. Evaluate and generate outputs

Advantages:

- Improves domain-specific performance
- Requires less data compared to training from scratch
- Saves time and computational resources

Limitations:

- Risk of overfitting
- Requires quality dataset
- Computational cost can still be high

3. Dataset for Story Generation

A dataset for creative story generation typically includes:

- Short stories
- Fictional narratives
- Fairy tales or prompts

The dataset should be:

- Clean and well-structured
- Rich in vocabulary and creativity
- Diverse in themes

4. Model Training Process

- The model learns patterns such as:
 - Sentence structure
 - Narrative flow
 - Character development
- Training is performed using loss functions like **Cross-Entropy Loss**.
- Optimization techniques such as **Adam optimizer** are used.

5. Text Generation

After fine-tuning, the model generates stories based on prompts.

Example Prompt:

“Once upon a time in a magical forest...”

Generated Output:

The model continues the story based on learned patterns.

Key Parameters:

- Temperature (controls randomness)
- Max length (controls story size)

- Top-k / Top-p sampling (controls diversity)

6. Evaluation of Model

Evaluation of creative text is subjective but can include:

- **Coherence:** Logical flow of story
- **Creativity:** Originality and imagination
- **Grammar:** Correct language usage
- **Relevance:** Alignment with prompt

7. Comparison: Base Model vs Fine-Tuned Model

Feature	Base GPT/GPT-2	Fine-Tuned Model
Creativity	Moderate	High
Coherence	Moderate	High
Domain Adaptation	Low	High
Relevance to Prompt	Moderate	High

8. Applications

- Story and content generation
- Chatbots with personality
- Game narrative design
- Script and screenplay writing
- Educational storytelling tools

Conclusion

Fine-tuning GPT/GPT-2 significantly enhances its ability to generate creative and meaningful stories. By training the model on a domain-specific dataset, it learns narrative structures, improves coherence, and produces more engaging content.

Although the base model provides general outputs, the fine-tuned model demonstrates better creativity, relevance, and storytelling ability. However, careful dataset selection and parameter tuning are essential to achieve optimal performance.

Overall, fine-tuning is a powerful technique for customizing language models for specialized applications such as creative writing, making AI-generated content more useful and human-like.